

Final Year Project Roadmap

Project Title

Early Detection of Malicious Ownership Transactions in Browser Extensions via Temporal Behavior Drift Modeling

First, Understand the Project in Simple Words

Your project is about detecting when a browser extension suddenly starts behaving differently.

For example:

- An extension normally asks for simple permissions.
- Later, after ownership changes or updates, it suddenly starts asking for dangerous permissions.
- It may start sending unusual network requests.
- It may access files, browser tabs, cookies, or APIs it never used before.

This sudden change in behavior is called behavioral drift.

Your goal is to build a system that: 1. Collects browser extension behavior data over time. 2. Learns what “normal” behavior looks like. 3. Detects unusual changes. 4. Sends an alert when something suspicious happens.

Complete Development Plan

Phase 1 — Learn Basic Concepts

Before coding, spend 1–2 weeks learning these topics:

Cybersecurity Concepts

- Browser extensions
- Chrome extension permissions
- Malware basics
- API abuse
- Network requests
- Cookies and local storage
- Privilege escalation

Machine Learning Concepts

- Supervised learning
- Unsupervised learning
- Anomaly detection
- Time series data
- Sequence modeling
- Classification metrics:
 - Accuracy
 - Precision
 - Recall
 - F1 Score
 - False Positive Rate

Python Concepts

- Variables and functions
 - Lists and dictionaries
 - Reading CSV files
 - Pandas
 - NumPy
 - Matplotlib
 - Basic machine learning with Scikit-learn
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Phase 2 — Install Required Tools

You should install these one by one.

Main Software

1. Python

Install Python 3.11 or above.

Website: <https://www.python.org/downloads/>

During installation:

- Check “Add Python to PATH”
- Install pip

2. VS Code

This will be your code editor.

Website: <https://code.visualstudio.com/>

Install these VS Code extensions:

- Python
- Jupyter
- Pylance
- GitLens

3. Git

You will use Git for version control.

Website: <https://git-scm.com/downloads>

4. Google Chrome

You will need Chrome because your project is based on browser extensions.

5. ChromeDriver

This is required for Selenium.

Website: <https://developer.chrome.com/docs/chromedriver/>

Important: Your ChromeDriver version should match your Chrome browser version.

Phase 3 — Install Python Libraries

Open terminal in VS Code and run:

```
pip install pandas numpy matplotlib seaborn scikit-learn jupyter selenium  
requests beautifulsoup4
```

Later, for machine learning and deep learning:

```
pip install tensorflow keras torch torchvision
```

For anomaly detection:

```
pip install pyod
```

For browser behavior logging:

```
pip install browsermob-proxy
```

Phase 4 — Suggested Project Folder Structure

```
FYP/  
|  
├─ data/  
|   ├─ raw_extensions/  
|   ├─ logs/  
|   └─ processed/  
|  
├─ notebooks/  
|   ├─ data_analysis.ipynb  
|   └─ model_training.ipynb  
|  
├─ scripts/  
|   ├─ collect_data.py  
|   ├─ preprocess.py  
|   ├─ train_model.py  
|   ├─ detect_drift.py  
|   └─ alert_system.py  
|  
├─ models/  
|   └─ trained_model.pkl  
|  
├─ reports/  
|   ├─ proposal.docx  
|   └─ final_report.docx  
|  
└─ README.md
```

Phase 5 — Decide What Data You Will Collect

You need behavioral data from browser extensions.

Examples:

- Permission changes
- API calls
- Network requests
- File access
- Tab access
- Cookie access
- Frequency of actions
- Extension updates
- Time of activity

Example row of data:

```
Extension Name: Adblock
Permissions: tabs, storage
Network Requests: 12
Cookie Access: No
API Calls: 25
Time: 10:30 PM
```

Phase 6 — How You Can Collect Data

You have several options.

Option 1 — Use Existing Public Datasets

This is easiest for FYP.

Possible sources:

- Chrome Web Store extension metadata
- GitHub browser extension repositories
- Malware datasets
- Kaggle cybersecurity datasets
- CRXcavator reports
- ExtensionTotal

Option 2 — Build Your Own Small Dataset

You can:

- Install some Chrome extensions

- Use Selenium to interact with them
 - Monitor behavior changes
 - Save logs in CSV format
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Phase 7 — Data Preprocessing

After collecting data:

- Remove missing values
- Convert text data into numbers
- Normalize values
- Convert timestamps into usable features
- Create labels:
 - Normal
 - Suspicious

Example:

```
Permission Count = 2
Network Requests = 15
Cookie Access = 0
Label = Normal
```

Phase 8 — Machine Learning Models You Can Use

Start with simple models first.

Beginner-Friendly Models

- Isolation Forest
- One-Class SVM
- Random Forest
- Logistic Regression

Advanced Models

- LSTM
- Autoencoders
- GRU
- Transformer-based anomaly detection

For your FYP, a very good combination is:

- Isolation Forest for anomaly detection
- LSTM for temporal behavior sequence modeling

This will make your project look stronger.

Phase 9 — Drift Detection Logic

Your project's core idea is:

If an extension's behavior suddenly changes beyond a threshold, raise an alert.

Example:

- Week 1:
 - Permissions = 2
 - Network Requests = 5
- Week 2:
 - Permissions = 8
 - Network Requests = 50
 - Cookie Access = Yes

This is suspicious behavioral drift.

Phase 10 — Alert System

When suspicious behavior is detected:

- Print warning in terminal
- Save alert in a CSV file
- Show alert in dashboard
- Send email notification

Example alert:

Warning: Extension 'XYZ Blocker' shows abnormal permission increase and network activity.

Phase 11 — Dashboard (Optional but Very Good)

You can make a simple dashboard using:

- Streamlit
- Flask
- Django

Best choice for beginners:

```
pip install streamlit
```

Then create a dashboard to show:

- Extension name
 - Current behavior
 - Risk score
 - Alerts
 - Charts
-

Phase 12 — Evaluation Metrics

You will measure:

- Accuracy
- Precision
- Recall
- F1 Score
- False Positive Rate
- ROC Curve

These are important for your report and viva.

Best Tech Stack Recommendation

- Python
- Pandas
- NumPy
- Scikit-learn
- TensorFlow or PyTorch
- Selenium

- Streamlit
 - CSV / SQLite
 - Matplotlib
-

Suggested Semester Plan

Semester 7

- Learn basics
- Install tools
- Collect data
- Preprocess data
- Train initial model

Semester 8

- Improve model
 - Add drift detection
 - Build dashboard
 - Add alert system
 - Test project
 - Write report
 - Prepare presentation
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What You Should Do This Week

1. Install Python, VS Code, Git, ChromeDriver.
 2. Learn basic Python and Pandas.
 3. Create your project folder structure.
 4. Install required libraries.
 5. Read about Chrome extension permissions.
 6. Learn Selenium basics.
 7. Start finding datasets.
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Important Advice

Do not try to build the entire project at once.

First make a very small working version:

- Collect extension behavior
- Save data
- Detect simple anomalies

Then slowly improve it.

That is the easiest and safest way to complete your FYP successfully.